SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Machine Learning: Supervised, Unsupervised & Ensemble Algorithms

Department: CSE AIML	**Course / Subject:** Machine Learning (AI501PC)
Semester: Semester 5	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Linear and Logistic Regression, Gradient Descent optimization, Decision Trees, Random Forests, Support Vector Machines (SVM), K-Means clustering, and Evaluation Metrics.

Unit 1: Supervised Regression

Simple & Multiple Linear Regression, Cost functions, Gradient Descent (Batch, Mini-batch, SGD), Polynomial Regression, Regularization (Ridge L2, Lasso L1).

Unit 2: Classification Models

Logistic regression, Decision Boundary, ROC-AUC curve, Confusion Matrix, Precision-Recall tradeoff, Naive Bayes classifier, and K-Nearest Neighbors (KNN).

Unit 3: Support Vector Machines & Kernels

Optimal separating hyperplanes, Hard vs Soft margin SVM, Slack variables, Kernel trick (Polynomial, Radial Basis Function RBF, Sigmoid).

Unit 4: Ensemble Learning

Bagging (Random Forests), Boosting (AdaBoost, Gradient Boosting, XGBoost), Voting Classifiers, Stacking, and Bias-Variance tradeoff analysis.

Unit 5: Unsupervised Learning

K-Means clustering algorithm, Elbow method, Hierarchical clustering (Dendrograms), DBSCAN density clustering, and Principal Component Analysis (PCA).